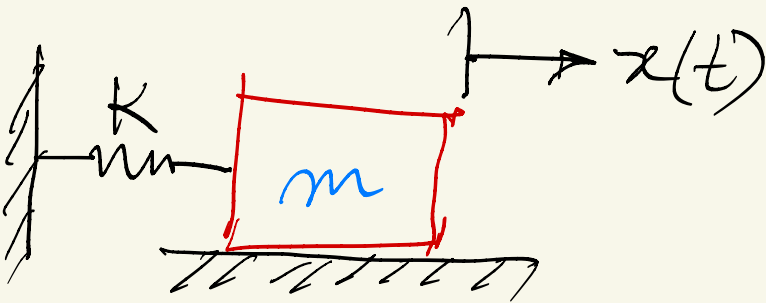




Lecture 2 :

Today we discussed how to form a differential equation of a system and how to find a solution of a differential equations.



$$\sum F_x = ma$$

$$\Rightarrow -kx = m \frac{d^2 x(t)}{dt^2}$$

$$\Rightarrow m \frac{d^2 x(t)}{dt^2} + kx = 0$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} + \frac{k}{m} x = 0$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} + \omega_n^2 x(t) = 0$$

$$\omega_n = \sqrt{k/m}$$

$$\omega_n^2 = \frac{k}{m} = \text{natural frequency}$$

$$x(t) = ??$$

Assume $x(t) = \cos \omega_n t$

differentiate $x(t)$

$$\frac{dx(t)}{dt} = -\omega_n \sin \omega_n t$$

$$\begin{aligned} \Rightarrow \frac{d^2 x(t)}{dt^2} &= -\omega_n^2 \cos \omega_n t \\ &= -\omega_n^2 x(t) \end{aligned}$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} + \omega_n^2 x(t) = 0$$

As a result, $x(t) = \cos \omega t$ is a solution of the differential equation.

Are there other solutions?

Yes

Infinitely many solutions exists.

How about $x(t) = \sin \omega t$

$$x(t) = e^{i\omega t}$$

$$x(t) = \sin \omega t + \cos \omega t$$

$$x(t) = e^{i\omega_n t} \quad i = \text{complex number}$$

$$\Rightarrow \frac{dx(t)}{dt} = i\omega_n e^{i\omega_n t}$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} = i^2 \omega_n^2 e^{i\omega_n t}$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} = -\omega_n^2 e^{i\omega_n t}$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} = -\omega_n^2 x(t)$$

$$\Rightarrow \frac{d^2 x(t)}{dt^2} + \omega_n^2 x(t) = 0$$

so $x(t) = e^{i\omega_n t}$ is a solution

Helpful Relations:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$